

DEVELOPMENT OF A DENTAL CARIES DIAGNOSIS SYSTEM USING DENTAL X-RAY IMAGES

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Abstract. Dental caries is one of the most prevalent oral diseases worldwide and significantly affects both health and quality of life. In dental practice, radiographic images play an important role in detecting carious lesions, especially in areas that are difficult to observe through clinical examination alone. However, radiographic interpretation still depends heavily on the experience of dentists and may result in variations in diagnostic outcomes. In recent years, Artificial Intelligence (AI) and Deep Learning have demonstrated considerable potential in supporting the detection and diagnosis of dental caries from radiographic images (Prados-Privado et al., 2020; Schwendicke et al., 2021; Kühnisch et al., 2022). This study proposes a deep learning-based system for dental caries detection and diagnosis using dental radiographs. The system consists of three stages: YOLOv8m is employed to detect suspicious carious regions, Region of Interest (ROI) extraction is performed, U-Net ResNet34 is utilized for lesion segmentation, and DenseNet121 is used to verify the diagnostic results. The integration of multiple models within a unified pipeline leverages the strengths of each approach, thereby improving the overall accuracy and reliability of the system. The dataset used in this study was constructed from the Caries Computer Vision Dataset available on the Roboflow platform. From the original dataset containing 9,344 dental radiographs, approximately 4,000 images were selected for training, validation, and evaluation. Experimental results indicate that the YOLOv8m model achieved a Precision of 86.65%, Recall of 72.30%, F1-score of 78.83%, and mAP@0.5 of 76.23%. The U-Net ResNet34 model obtained a Dice Score of 84.76% and an IoU of 73.96%, while DenseNet121 achieved an Accuracy of 96.88% and an F1-score of 96.84%. These results demonstrate that the combination of YOLOv8m, U-Net ResNet34, and DenseNet121 can effectively support dental caries detection and diagnosis from dental radiographic images. The study highlights the potential of Deep Learning applications in dentistry and provides a foundation for the development of automated diagnostic support systems in the future.

Keywords: Dental Caries, Dental Radiographs, Deep Learning, YOLOv8m, U-Net ResNet34, DenseNet121

1 Introduction

Dental caries is one of the most common oral diseases worldwide and remains a major public health concern. According to the World Health Organization (WHO), oral diseases affect approximately 3.5 billion people globally, with permanent dental caries being the most prevalent condition. If not detected and treated promptly, dental caries can lead to pain, pulp inflammation, tooth loss, and a significant reduction in quality of life (World Health Organization, 2025). Furthermore, studies on the global burden of disease indicate that the number of individuals affected by dental caries continues to increase and is expected to rise further in the coming years, placing substantial pressure on oral healthcare systems worldwide (Li et al., 2025). To improve global oral health, the FDI World Dental Federation introduced the Vision 2030 strategy, emphasizing the role of digital technologies and artificial intelligence in enhancing access to high-quality dental care services (FDI World Dental Federation, 2021).

In Vietnam, dental caries remains highly prevalent among children and adolescents. A study conducted by Chuyen et al. (2021) among secondary school students in mountainous regions reported a high prevalence of dental caries, influenced by oral hygiene habits and healthcare accessibility. These findings highlight the urgent need for effective solutions that support early detection and accurate diagnosis of dental caries. In dental practice, radiographic imaging plays a critical role in identifying lesions that may be difficult to detect through clinical examination. Imaging modalities such as Bitewing and Panoramic radiographs enable dentists to evaluate the location, size, and extent of carious lesions. However, radiographic interpretation remains highly dependent on the clinician's expertise, which may lead to inconsistencies in diagnosis among practitioners.

Recent advances in Artificial Intelligence (AI) and Deep Learning have achieved remarkable success in medical image analysis. Deep learning models can automatically learn discriminative features from image data and have demonstrated outstanding performance in object detection, image classification, and image segmentation tasks. Numerous studies have shown that AI-based approaches can assist in detecting dental caries with high accuracy, reducing the workload of dental professionals and improving diagnostic consistency (Prados-Privado et al., 2020; Schwendicke et al., 2021; Kühnisch et al., 2022). To address these challenges, this study proposes an intelligent dental caries diagnosis system based on a multi-stage deep learning framework. The proposed system employs YOLOv8m for lesion detection, ROI extraction for focusing on suspicious regions, U-Net ResNet34 for lesion segmentation, and DenseNet121 for final lesion verification. By combining detection, segmentation, and classification within a unified workflow, the system aims to improve diagnostic accuracy and reliability while providing effective support for clinical dental practice.

2 THEORETICAL BACKGROUND AND RELATED WORKS

In recent years, Artificial Intelligence (AI) and Deep Learning have become prominent research directions in the field of medical image analysis. Owing to their ability to automatically learn features from image data, deep learning models have been widely applied to various disease diagnosis support tasks, helping improve accuracy and consistency compared with manual assessment methods. In dentistry, AI has been utilized to support the detection, classification, and segmentation of dental caries lesions on radiographic images, thereby enhancing diagnostic efficiency and reducing dependence on the subjective experience of clinicians (Prados-Privado et al., 2020; Schwendicke et al., 2021; Kühnisch et al., 2022).

Before deep learning models became widely adopted, many studies employed traditional machine learning approaches for data classification problems. Ho et al. (2013) proposed a One-Class Support Vector Machine (One-Class SVM) approach for environments with evolving data distributions. Subsequent studies by Ho (2015), Ho et al. (2016), and Ho et al. (2017) developed dynamic classification and incremental learning models to improve adaptability to new data. Although these approaches achieved promising results, they relied heavily on handcrafted feature extraction and encountered limitations when processing complex image data. The emergence of Deep Learning introduced a more effective approach for medical image analysis tasks.

In addition to traditional machine learning methods, numerous studies have focused on developing deep learning models to improve the performance of dental image analysis. Prior to the widespread adoption of modern deep learning architectures, several studies employed dynamic classification and incremental learning approaches based on One-Class SVM to handle evolving data streams. Research conducted by Ho and colleagues (2013, 2016, 2017) demonstrated the adaptability of classification models in unstable data environments. However, these methods depended significantly on manual feature engineering and faced challenges when dealing with complex image datasets. Deep Learning introduced a new paradigm with the capability to automatically learn representative features directly from input data.

Current studies primarily focus on three major approaches: lesion detection, dental caries classification, and lesion segmentation.

For dental caries detection, deep learning models have demonstrated the ability to accurately identify lesion locations on dental radiographs. Srivastava et al. (2017) were among the earliest researchers to apply Deep Learning to dental caries detection on bitewing radiographs. Subsequently, numerous studies further developed deep learning-based systems for caries detection on bitewing images, including those by Cantu et al. (2020), Chen et al. (2022), García-Cañas et al. (2022), Estai et al. (2022), Pérez de Frutos et al. (2024), and Tan et al. (2025). Their findings indicate that deep learning models can effectively support lesion detection with high accuracy and strong potential for clinical applications.

For classification tasks, Convolutional Neural Networks (CNNs) have been extensively applied to determine the presence or severity of dental caries in radiographic

images. The study by Cantu et al. (2020) demonstrated that deep CNN models were capable of detecting different stages of dental caries on bitewing radiographs with promising accuracy. Panyarak et al. (2023) applied deep learning techniques to classify dental caries severity according to the International Caries Classification and Management System (ICCMS™) and achieved favorable results on bitewing datasets. More recently, Kim et al. (2025) combined CNNs with Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize image regions influencing model decisions, thereby improving the interpretability of diagnostic outcomes. Furthermore, the study by ForouzeshFar et al. (2024) continued to confirm the effectiveness of CNN-based architectures for dental caries diagnosis using dental radiographs.

DenseNet121 is one of the modern CNN architectures widely adopted in medical image classification tasks. This architecture was proposed by Huang et al. (2017) and employs dense connections between network layers, enabling efficient feature reuse, reducing information loss, and improving representation learning capability. Owing to these advantages, DenseNet121 was selected in this study to verify diagnostic results on suspicious carious regions.

For image segmentation, U-Net is among the most widely used architectures in medical image analysis. U-Net was proposed by Ronneberger et al. (2015) with an encoder–decoder structure that enables both feature extraction and preservation of spatial information. Subsequently, Oktay et al. (2018) introduced Attention U-Net by incorporating an attention mechanism that allows the model to focus more effectively on small lesion regions. Studies in dentistry have shown that U-Net-based architectures can accurately identify lesion boundaries on radiographic images, thereby supporting disease assessment and monitoring processes.

Table 1. Summary of Related Studies

Study	Category	Main Metrics
Pérez de Frutos et al. (2024)	<i>Detection</i>	<i>F1-score = 0.548</i>
Marginean et al. (2024)	<i>Segmentation</i>	<i>Dice = 0.682, IoU = 0.518</i>
Helli & Hamamci (2022)	<i>Segmentation</i>	<i>IoU $\approx$ 0.61</i>
Cantu et al. (2020)	<i>Classification</i>	<i>Accuracy $\approx$ 0.80</i>
Panyarak et al. (2023)	<i>Classification</i>	<i>Accuracy $\approx$ 0.78</i>

Table 1 shows that current studies mainly focus on individual tasks such as detection, classification, or segmentation of dental caries in radiographic images. Detection studies have demonstrated the capability of deep learning models to identify lesion locations, whereas classification studies focus on confirming the presence or severity of lesions. Segmentation studies, particularly those based on U-Net architectures, enable precise delineation of lesion boundaries on radiographs. However, most existing studies address only a single task independently. Therefore, this study proposes an integrated framework combining YOLOv8m, U-Net ResNet34, and DenseNet121 to simultaneously perform detection, segmentation, and classification within a unified processing pipeline, thereby improving diagnostic accuracy and reliability.

Based on the reviewed literature, it can be observed that most existing solutions focus on a single task, such as detection, classification, or segmentation. Integrated systems capable of performing multiple tasks within the same processing workflow remain relatively limited. Therefore, this study proposes the development of a Deep Learning-based dental caries diagnosis support system for dental radiographs. The proposed workflow employs YOLOv8m to detect suspicious carious regions, U-Net ResNet34 to segment lesion areas, and DenseNet121 to verify diagnostic results. This approach aims to leverage the strengths of each model to improve diagnostic accuracy, reliability, and clinical decision support for dental caries detection and assessment.

3 PROPOSED DENTAL CARIES DIAGNOSIS SYSTEM

In this study, a dental caries diagnosis support system for dental radiographs was developed based on the integration of three deep learning models: YOLOv8m, U-Net ResNet34, and DenseNet121. This approach builds upon previous research directions involving lesion detection using object detection models, lesion segmentation using U-Net-based architectures, and image classification using modern convolutional neural networks (Tan et al., 2025; Pérez de Frutos et al., 2024; Ronneberger et al., 2015; Huang et al., 2017).

The objective of the proposed system is to automatically detect suspicious carious regions, accurately determine lesion boundaries, and verify diagnostic results in order to improve the accuracy and reliability of the diagnosis process.

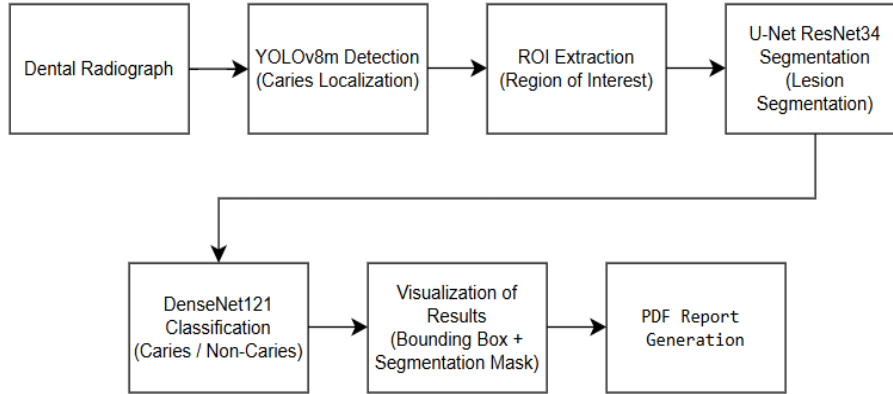


Fig. 1. Proposed Dental Caries Diagnosis System Architecture (This figure presents the end-to-end workflow of the proposed system: Dental Radiograph → YOLOv8m Detection → ROI Extraction → U-Net ResNet34 Segmentation → DenseNet121 Classification → Result Visualization → PDF Report Generation.)

YOLOv8m is first employed to detect suspicious carious regions on dental radiographs. After lesion detection, the system extracts Regions of Interest (ROIs) to re-

move irrelevant areas and focus further analysis on suspected lesion regions. The extracted ROIs are then fed into the U-Net ResNet34 model to perform lesion segmentation. Subsequently, the segmentation results are passed to DenseNet121 to verify the presence of dental caries and reduce false-positive predictions. Finally, the diagnostic results are displayed through a web-based interface and exported as PDF reports for storage and follow-up purposes.

The dataset used in this study was collected from Roboflow Universe under the name Caries Computer Vision Dataset. The original dataset contains approximately 9,344 annotated dental radiographs for dental caries detection and segmentation tasks.

Due to training time and computational resource constraints, approximately 4,000 images were selected from the original dataset to construct the experimental dataset. These images were reviewed, preprocessed, and used for model training, validation, and evaluation.

Table 2. Summary of Related Studies

Dataset Split	Number of Images
Training	3,200
Validation	500
Test	300
Total	4,000

Before training, several preprocessing steps were applied to improve the quality of the input data. First, all radiographic images were resized according to the requirements of each model. Subsequently, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied to enhance image contrast and highlight low-contrast lesion regions in dental radiographs.

After YOLOv8m detects suspicious carious regions, the system extracts Regions of Interest (ROIs) from the predicted bounding boxes. To avoid information loss at lesion boundaries, ROI regions are expanded using a padding technique before cropping. The extracted ROIs are stored separately and subsequently used for segmentation and classification model training.

The use of ROIs helps reduce noise from irrelevant anatomical structures in radiographic images and enables the models to focus on suspicious lesion areas, thereby improving processing efficiency and reducing computational costs.

YOLOv8m was employed to detect suspicious dental caries regions in dental radiographs. YOLOv8m belongs to the YOLO (You Only Look Once) family of object detection models and has demonstrated excellent real-time object detection performance across various computer vision applications. Recent studies have shown that Deep Learning-based detection models can effectively support dental caries detection on bitewing radiographs (Pérez de Frutos et al., 2024; Tan et al., 2025).

In this study, YOLOv8m was trained using a dataset annotated with bounding boxes containing a single object class, namely Caries.

Table 3. YOLOv8m Training Parameters

Parameter	Value
Epochs	25
Image Size	800
Batch Size	4
Optimizer	<i>AdamW</i>
Learning Rate	<i>0.00003</i>

After suspicious lesion regions were detected, the extracted ROIs were fed into the U-Net ResNet34 model for lesion segmentation. U-Net is one of the most widely used medical image segmentation architectures due to its capability of combining semantic and spatial information effectively (Ronneberger et al., 2015).

Furthermore, several studies in dentistry have demonstrated the effectiveness of U-Net-based architectures for tooth and lesion segmentation on dental radiographs (Helli & Hamamcı, 2022; Marginean et al., 2024).

In this study, ResNet34 was employed as the encoder backbone to improve feature extraction capability and segmentation quality. The loss function was designed as a combination of Binary Cross-Entropy Loss and Dice Loss to optimize segmentation performance, particularly for small lesion regions.

Table 4. U-Net ResNet34 Training Parameters

Parameter	Value
Epochs	24
Image Size	512
Batch Size	8
Optimizer	<i>AdamW</i>
Learning Rate	<i>0.00001</i>

Following segmentation, the lesion regions were passed to DenseNet121 for diagnostic verification. DenseNet121 is a modern CNN architecture proposed by Huang et al. (2017), utilizing dense connections between network layers to improve feature reuse and mitigate information loss during data propagation.

The model was initialized with ImageNet pretrained weights and trained using Transfer Learning techniques to leverage previously learned image features. DenseNet121 performed a binary classification task consisting of two classes: Caries and Non-Caries.

Table 5. DenseNet121 Training Parameters

Parameter	Value
Epochs	10
Image Size	224
Batch Size	32

Optimizer	<i>Adam</i>
Learning Rate	<i>0.00001</i>

After training was completed, the best-performing models were saved and integrated into a web-based system developed using Flask. The system allows users to upload dental radiographs, perform automatic analysis, and visualize diagnostic results.

The output includes lesion locations detected by YOLOv8m, lesion masks generated by U-Net ResNet34, and verification results produced by DenseNet121. In addition, the system supports exporting diagnostic reports in PDF format for documentation, storage, and future reference.

The integration of YOLOv8m, U-Net ResNet34, and DenseNet121 within a unified processing pipeline enables the system to leverage the strengths of detection, segmentation, and classification approaches simultaneously, thereby improving the accuracy and reliability of the proposed dental caries diagnosis support system.

4 EXPERIMENTS AND EVALUATION

The models in this study were trained and evaluated on the Google Colab platform using an NVIDIA Tesla T4 Graphics Processing Unit (GPU). The system was developed using the Python programming language together with widely used deep learning and image processing libraries.

Table 6. Experimental Environment

Component	Configuration
Platform	<i>Google Colab</i>
GPU	<i>Tesla T4</i>
Programming Language	<i>Python</i>
Detection Model	<i>YOLOv8m</i>
Segmentation Model	<i>U-Net ResNet34</i>
Classification Model	<i>DenseNet121</i>
Web Framework	<i>Flask</i>

To evaluate model performance, several metrics were employed, including Precision, Recall, F1-score, Accuracy, Intersection over Union (IoU), Dice Score, and Mean Average Precision (mAP).

For the detection task, Precision, Recall, F1-score, and mAP were used.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

For the classification task, Accuracy was used as the primary evaluation metric.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

For the segmentation task, IoU and Dice Score were employed.

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|}$$

$$\text{Dice Score} = \frac{2|A \cap B|}{|A| + |B|}$$

YOLOv8m was utilized to detect suspicious dental caries regions in dental radiographs. Model performance was evaluated using Precision, Recall, F1-score, and mAP.

Table 7. YOLOv8m Results on the Test Set

Metric	Value
Precision	0.8665
Recall	0.7230
F1-score	0.7883
mAP@0.5	0.7623
mAP@0.5:0.95	0.4476

The results indicate that YOLOv8m achieved a Precision of 86.65%, demonstrating that most detected regions corresponded to actual carious lesions. The Recall of 72.30% indicates that the model was able to identify the majority of dental caries cases in the test dataset. Furthermore, the mAP@0.5 value of 76.23% reflects the model's ability to localize lesion regions with reasonable accuracy, making it suitable as the initial detection stage in the proposed system.

After suspicious lesion regions were detected by YOLOv8m, the system extracted ROIs and fed them into the U-Net ResNet34 model for lesion segmentation.

Table 7. U-Net ResNet34 Results on the Test Set

Metric	Value
Dice Score	0.8476
IoU	0.7396

The results demonstrate that U-Net ResNet34 can accurately identify the boundaries of dental caries lesions. The Dice Score of 84.76% indicates a high degree of similar-

ty between the predicted masks and the ground-truth annotations. In addition, the IoU value of 73.96% confirms that the model provides stable and reliable segmentation performance for the target task.

To improve system reliability, DenseNet121 was employed as the final verification stage to confirm suspicious regions after detection and segmentation.

Table 7. DenseNet121 Results on the Test Set

Metric	Value
Accuracy	<i>0.9688</i>
Precision	<i>0.9814</i>
Recall	<i>0.9558</i>
F1-score	<i>0.9684</i>

The results show that DenseNet121 achieved an Accuracy of 96.88%, Precision of 98.14%, Recall of 95.58%, and F1-score of 96.84%. These findings demonstrate the model’s capability to distinguish between carious and non-carious regions, thereby reducing false-positive cases and improving the reliability of the overall system.

To assess the effectiveness of the proposed solution, the obtained results were compared with several representative studies previously published on dental caries detection, classification, and segmentation using dental radiographs.

Table 10. Comparison with Previous Studies

Study	Detection	Segmentation	Classification
Panyarak et al. (2023)	-	-	<i>Accuracy ≈ 0.78</i>
Pérez de Frutos et al. (2024)	<i>F1-score = 0.548</i>	-	-
Marginean et al. (2024)	-	Dice = 0.682, IoU = 0.518	-
Proposed Study	<i>F1-score = 0.7883, mAP@0.5 = 0.7623</i>	Dice = 0.8476, IoU = 0.7396	<i>Accuracy = 0.9688, F1-score = 0.9684</i>

As shown in Table 10, the proposed system achieved promising results across all three tasks, including detection, segmentation, and classification. For the detection task, YOLOv8m achieved an F1-score of 0.7883, outperforming the F1-score of 0.548 reported by Pérez de Frutos et al. (2024). In the segmentation task, U-Net ResNet34 achieved a Dice Score of 0.8476 and an IoU of 0.7396, exceeding the performance reported by Marginean et al. (2024), which achieved a Dice Score of 0.682 and an IoU of 0.518. The proposed segmentation model also outperformed the results reported by Helli and Hamamcı (2022), which obtained Dice = 0.74 and IoU = 0.61.

For the classification task, DenseNet121 achieved an Accuracy of 96.88%, significantly higher than the approximately 78–80% Accuracy reported by Cantu et al. (2020) and Panyarak et al. (2023). These results indicate that integrating YOLOv8m, U-Net ResNet34, and DenseNet121 contributes to more effective support for dental caries diagnosis using dental radiographs.

Overall, combining YOLOv8m, U-Net ResNet34, and DenseNet121 within a unified processing pipeline enables the system to leverage the strengths of detection, segmentation, and classification approaches. As a result, the proposed system achieves higher accuracy and reliability than approaches relying on a single model.

Despite the promising results, several limitations remain. The dataset used in this study primarily focuses on dental caries and does not comprehensively cover other dental diseases. In addition, the models may still miss very small lesions or lesions with low contrast in radiographic images. Furthermore, the proposed system has only been evaluated on the research dataset and has not yet been validated using real-world clinical data collected from dental clinics and hospitals.

These limitations provide important directions for future research aimed at improving the generalization capability and practical applicability of the proposed dental caries diagnosis system.

5 CONCLUSION

In this study, a Deep Learning-based dental caries detection and diagnosis support system using dental radiographs was successfully developed. The proposed system consists of three main stages: YOLOv8m for detecting suspicious carious regions, U-Net ResNet34 for lesion segmentation, and DenseNet121 for verifying diagnostic results on the extracted Regions of Interest (ROIs). After training, the models were integrated into a web-based system developed using Flask, allowing users to upload dental radiographs, perform automatic analysis, and generate diagnostic reports in PDF format. The dataset used in this study was constructed from the Caries Computer Vision Dataset available on the Roboflow platform. Approximately 4,000 dental radiographs were selected from the original dataset for experimentation. The images were preprocessed using Contrast Limited Adaptive Histogram Equalization (CLAHE), and ROI datasets were generated to improve the quality of the input data for deep learning models. Experimental results demonstrated that YOLOv8m achieved a Precision of 86.65%, a Recall of 72.30%, and an mAP@0.5 of 76.23%. U-Net ResNet34 achieved a Dice Score of 84.76% and an IoU of 73.96%, while DenseNet121 achieved an Accuracy of 96.88% and an F1-score of 96.84%. These results indicate that the proposed system is capable of accurately detecting, segmenting, and verifying dental caries lesions from dental radiographs. The main contribution of this study is the proposed multi-stage ROI-based processing framework that integrates Detection, Segmentation, and Classification within a single system. This approach leverages the strengths of each model, thereby improving diagnostic reliability and reducing false-positive predictions compared with approaches that rely on a single model. Overall, the findings demonstrate the feasibility of applying YOLOv8m, U-Net ResNet34, and DenseNet121 to support dental caries diagnosis using dental radiographs. The proposed system has the potential to be applied in clinical decision-support solutions, assisting dentists in detecting and evaluating dental caries more quickly, visually, and consistently.

References

1. Cantu, A. G., Gehrung, S., Krois, J., Chaurasia, A., Rossi, J. G., Gaudin, R., Elhennawy, K., Schwendicke, F., & Paris, S. (2020). Detecting caries lesions of different radiographic extension on bitewings using deep learning. *Journal of Dentistry*, 100, 103425. <https://doi.org/10.1016/j.jdent.2020.103425>
2. Chen, Y., et al. (2022). Detection of proximal caries lesions on bitewing radiographs using deep learning method. *Caries Research*. <https://doi.org/10.1159/000527418>
3. Chuyen, N. V., Van Duong, H., Khanh, L. N., et al. (2021). The prevalence of dental caries and associated factors among secondary school children in rural highland Vietnam. *BMC Oral Health*, 21(1), 349. <https://doi.org/10.1186/s12903-021-01704-y>
4. Estai, M., Kanagasingam, Y., Huang, B., et al. (2022). Evaluation of a deep learning system for automatic detection of proximal surface dental caries on bitewing radiographs. *Oral Surgery, Oral Medicine, Oral Pathology and Oral Radiology*, 134(2), 220–227. <https://doi.org/10.1016/j.oooo.2022.03.008>
5. FDI World Dental Federation. (2021). Vision 2030: Delivering optimal oral health for all. <https://www.fdiworlddental.org/vision-2030-delivering-optimal-oral-health-all>
6. ForouzeshFar, P., Safaei, A. A., Ghaderi, F., & Hashemikamangar, S. S. (2024). Dental caries diagnosis from bitewing images using convolutional neural networks. *BMC Oral Health*, 24, 211. <https://doi.org/10.1186/s12903-024-03973-9>
7. García-Cañas, J., et al. (2022). Deep learning-based caries detection on bitewing radiographs. *Caries Research*. <https://doi.org/10.1159/000527491>
8. Helli, S., & Hamamcı, A. (2022). Tooth instance segmentation on panoramic dental radiographs using U-Nets and morphological processing. *Düzce University Journal of Science and Technology*, 10(1), 39–50. <https://doi.org/10.29130/dubited.950568>
9. Ho, A. K. N. (2015). Méthodes de classifications dynamiques et incrémentales: Application à la numérisation cognitive d'images de documents (Doctoral dissertation, University of Tours, France). https://www.rfai.lifat.univ-tours.fr/PublicData/PhD/these_vf_khoi.pdf
10. Ho, A. K. N., Eglin, V., Ragot, N., & Ramel, J.-Y. (2016). Multi one-class incremental SVM for document stream digitization. In *Proceedings of the 12th IAPR International Workshop on Document Analysis Systems (DAS 2016)*. <https://hal.science/hal-01307021v1/document>
11. Ho, A. K. N., Eglin, V., Ragot, N., & Ramel, J.-Y. (2017). A multi-one-class dynamic classifier for adaptive digitization of document streams. *International Journal on Document Analysis and Recognition*, 20(3), 137–154. <https://hal.science/hal-01525831v1>
12. Ho, A. K. N., Ragot, N., Ramel, J. Y., Eglin, V., & Sidere, N. (2013). Document classification in a non-stationary environment: A one-class SVM approach. In *2013 12th International Conference on Document Analysis and Recognition (ICDAR)* (pp. 616–620). <https://doi.org/10.1109/ICDAR.2013.127>
13. Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 4700–4708). <https://doi.org/10.1109/CVPR.2017.243>
14. Kim, D., Kim, J., & Choi, S. G. (2025). CNN-based remote dental diagnosis model for caries detection with Grad-CAM. *Scientific Reports*, 15, 26555. <https://doi.org/10.1038/s41598-025-11447-3>
15. Kühnisch, J., Meyer, O., et al. (2022). Performance of artificial intelligence for detection and classification of dental caries: A meta-analysis. *Clinical Oral Investigations*, 26, 1313–1327. <https://doi.org/10.1007/s00784-021-04327-3>

16. Li, X., Li, R., et al. (2025). Global burden of dental caries from 1990 to 2021 and future projections. *International Dental Journal*, 75(5), 100904. <https://doi.org/10.1016/j.identj.2025.100904>
17. Marginean, A. C., Mureşanu, S., Hedeşiu, M., & Dioşan, L. (2024). Teeth segmentation and carious lesions segmentation in panoramic X-ray images using CariSeg, a networks' ensemble. *Heliyon*, 10(10), e30836. <https://doi.org/10.1016/j.heliyon.2024.e30836>
18. Oktay, O., Schlemper, J., Folgoc, L. L., et al. (2018). Attention U-Net: Learning where to look for the pancreas. *arXiv*. <https://arxiv.org/abs/1804.03999>
19. Panyarak, W., Wantanajittikul, K., Suttapak, W., Charuakkra, A., & Prapayasatok, S. (2023). Feasibility of deep learning for dental caries classification in bitewing radiographs based on the ICCMS™ radiographic scoring system. *Oral Surgery, Oral Medicine, Oral Pathology and Oral Radiology*, 135(2), 272–281. <https://doi.org/10.1016/j.oooo.2022.06.012>
20. Pérez de Frutos, J., et al. (2024). AI-Dentify: Deep learning for proximal caries detection on bitewing radiographs. *BMC Oral Health*, 24, 1098. <https://doi.org/10.1186/s12903-024-04120-0>
21. Prados-Privado, M., García Villalón, J., Martínez-Martínez, C. H., Ivorra, C., & Prados-Frutos, J. C. (2020). Dental caries diagnosis and detection using neural networks: A systematic review. *Journal of Clinical Medicine*, 9(11), 3579. <https://doi.org/10.3390/jcm9113579>
22. Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI 2015)* (pp. 234–241). https://doi.org/10.1007/978-3-319-24574-4_28
23. Schwendicke, F., Samek, W., & Krois, J. (2021). Artificial intelligence in dental diagnostics: A review. *Journal of Dental Research*, 100(7), 769–778. <https://doi.org/10.1177/0022034521993885>
24. Srivastava, M. M., Kumar, P., Pradhan, L., & Varadarajan, S. (2017). Detection of tooth caries in bitewing radiographs using deep learning. *arXiv*. <https://arxiv.org/abs/1711.07312>
25. Tan, Z. Q., et al. (2025). Caries detection in primary molars with bitewing radiographs through deep learning-based object detectors. *Caries Research*. <https://doi.org/10.1159/000550079>
26. World Health Organization. (2025). Oral health. <https://www.who.int/news-room/fact-sheets/detail/oral-health>